

Problem Statement

You are given a grid of dimensions $N \times M$. Each cell contains one of the following values:

0 → Empty cell 1 → Fresh orange 2 → Rotten orange

A rotten orange can rot an adjacent fresh orange in 1 unit of time.

A rotten orange at position (i, j) can rot fresh oranges at:

$(i-1, j)$ → Up $(i+1, j)$ → Down $(i, j-1)$ → Left $(i, j+1)$ → Right

Find the minimum time required to rot all fresh oranges.

If it is impossible to rot all fresh oranges, print -1.

If there are no fresh oranges initially, the required time is 0.

Input Format

The first line contains an integer T , the number of test cases. For each test case: The first line contains two integers N and M , representing the number of rows and columns. The next N lines contain M space-separated integers representing the grid.

Constraints

$1 \leq T \leq 100$ $1 \leq N, M \leq 500$ $\text{grid}[i][j] \in \{0, 1, 2\}$

Output Format

Print a single integer representing the minimum time required

Change Theme

Language Python 3



```

8  fresh=[]
9  for i in range(n):
10     for j in range(m):
11         if oranges[i][j]==2:
12             rotten.append((i,j))
13         if oranges[i][j]==1:
14             fresh.append((i,j))
15  c=0
16  while fresh and rotten:
17     new_rot=[]
18     for i,j in rotten:
19         for n_i,n_j in [(i+1,j),(i-1,j),(i,j+1),(i,j-1)]:
20             if (n_i,n_j) in fresh:
21                 fresh.remove((n_i,n_j))
22                 new_rot.append((n_i,n_j))
23     c+=1
24     if not fresh:
25         break
26     rotten=new_rot
27  print(c if not fresh else -1)
28

```

Line: 28 Col: 1



Upload Code as File

Run Code

Submit Code



Test against custom input